

MODEL FORMULATION

I-Star is a cost allocation approach where participants incur costs based on the size of their order and the overall participation with market volumes. The idea behind the model follows from economic supply-demand equilibrium starting at the total cost level¹. The model is broken down into two components: Instantaneous Impact denoted as I-Star or I* and Market Impact which denoted as MI which represents impact cost due to the specified trading strategy. This impact function is broken down into a temporary and permanent term.

I-Star: Instantaneous Impact Equation

$$I_{bp}^* = a_1 \cdot \left(\frac{Q}{ADV} \right)^{a_2} \cdot \sigma^{a_3} \quad (4.6)$$

In trading, I-Star represents what we call theoretical instantaneous impact cost incurred by the investor if all shares were released to the market. This component can also be thought of as the total payment required to attract additional sellers or buyers to the marketplace. For example, the

¹The reasoning behind this formulation and how it diverges from the Almgren & Chriss expression is simple. Prior to moving into the financial industry I was employed by R.J. Rudden Associates, Inc., a leading global consulting firm specializing in utility cost of service studies as part of rate cases. In these cases, utilities (both natural gas and electric companies) formulated studies to determine serving cost per customer class by mapping actual costs to usage point and allocating this quantity to each party based on usage percentage. This would ensure each customer paid only for services consumed based on the cost of providing that service. While many services were easy to compute usage by each customer, such as electric kwh or natural btu consumption, others were more difficult to compute due to services or mechanics that are shared across customers. For example, there are multiple parties sharing the same generators, overhead transmission lines, pipeline and natural gas storage facilities, as well as corporate functions such as strategy and administrative services. The basic concept of these studies was that we started with a total cost value that was known from accounting records, and then these costs were mapped and allocated to the appropriate customer based on usage and cost to provide the service. This was done to ensure a fair and equitable system across all customers so that no single customer class was being charged more than their fair usage. Those who played a large role (and unknowingly) in the development of an industry leading market impact model include: Rich Rudden, Steve Maron, John Little, Russ Feingold, Kevin Harper, and William Hederman. Thus fittingly, when I was presented with a project to compute and estimate market impact cost, the modeling approach I undertook followed this cost allocation methodology. The I-Star model follows directly from this system. Actual costs as mapped to their underlying components and allocated to point of usage. The methodology is described in this chapter and as we show has many appealing properties for execution strategies, algorithmic trading rules, as well as for portfolio optimization and basket trading strategies.